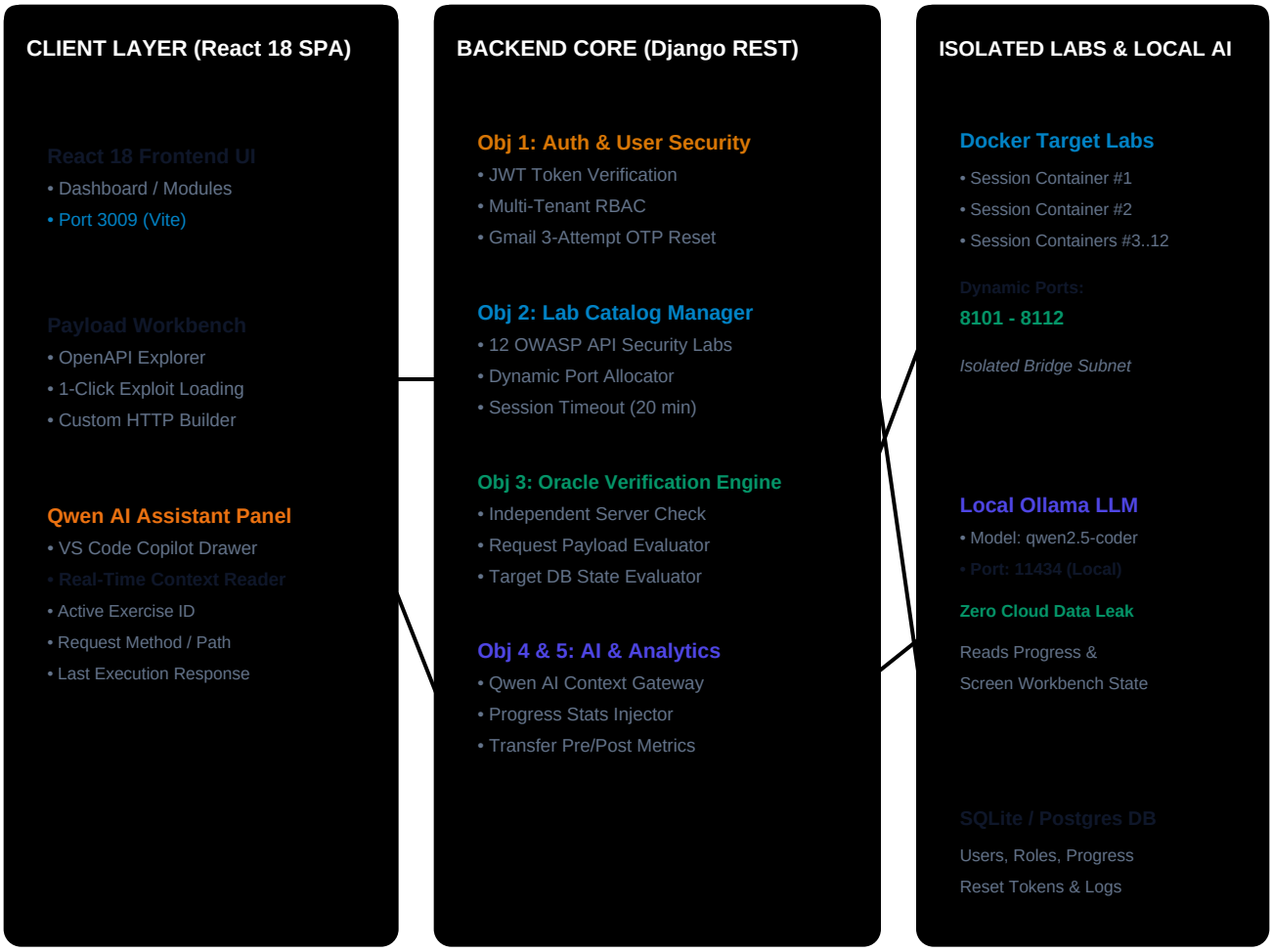
High-Level System Architecture Specification

HackTheAPI // OWASP API Security Training & Assessment Platform — All 5 Core Objectives Integrated



System Architecture Component & Objective Mapping

Component Layer	Target Technology / Port	System Objective & Responsibility
1. Client UI Layer	React 18 SPA (Vite) Port 3009	Renders interactive dashboard, 12 OWASP payload workbenches, OpenAPI explorer, and VS Code Copilot style Qwen AI drawer.
2. Auth & RBAC Core	Django REST Framework Port 8000	**Objective 1:** Multi-tenant RBAC (Student, Instructor, Assessor, Admin) with Gmail SMTP 3-attempt OTP password reset security.
3. OWASP Labs Catalog	Docker Target Containers Ports 8101 - 8112	**Objective 2:** 12 hands-on vulnerability lab containers running on isolated per-user bridge subnets.
4. Server-Side Oracle Engine	Django Evaluation Core Port 8000	**Objective 3:** Independent dual-phase evaluator asserting HTTP request/response payloads and physical backend container database states.

5. Transfer Analytics Engine	Django Analytics Core Port 8000	**Objective 4:** Pre-test vs Post-test transfer learning assessment engine tracking skill acquisition velocity and student progress metrics.
6. Local Ollama AI Engine	Ollama (`qwen2.5-coder`) Port 11434 (Localhost)	**Objective 5:** Zero cloud data leak local LLM service reading user progress and real-time screen/workbench context for step-by-step lab guidance.